Three ways to represent a model (answer key)

Every linear regression model can be represented three ways. On each of the following pages you'll find an environmental case study and **one** representation of its model. Your job is to create the other **two**.

- **Equation.** The model in statistical notation, with values for each coefficient. Write categorical predictors as dummy variables and say what each one means. Include the standard deviation of the residuals: $\varepsilon \sim \text{Normal}(0, \sigma)$.
- **Visual.** A scatter plot of the response and predictor(s), with the regression line(s) drawn on top. Label the axes and add tick labels. For a categorical predictor, draw one line (or one group mean) per level. For an interaction between two continuous predictors, draw lines for two or three values of the second predictor.
- **Computational.** The output of `summary()` for the `lm()` model in R. Write the formula, the coefficient names as R would print them (for example, `mountTracker` or `urchins:protectionMPA`), and the estimates.

Your three representations should describe the same model. Use the variable names, units, and reference levels in each page's table. Sketches don't need to be exact, but the intercepts and slopes should be reasonable.

When you finish a page, check your work: set the continuous predictor to 0 and use the reference level. Does your equation give the same value as the intercept in your `summary()` output and the reference line in your figure?

	Model	**Case study**	**Given**
1	One continuous predictor	Restoration	Equation
2	One categorical predictor (2 levels)	Groundwater	Visual
3	One categorical predictor (3 levels)	Wildfire	Computational
4	One continuous + one categorical (2 levels)	Green energy	Computational
5	One continuous + one categorical (3 levels)	Climate	Visual
6	One continuous interacting with one categorical (2 levels)	Marine and coastal	Visual
7	One continuous interacting with one continuous	Forestry	Equation

1. Restoring coastal sage scrub

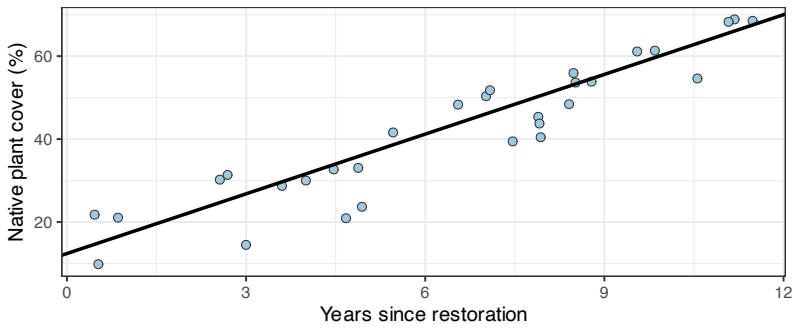
A land trust has replanted coastal sage scrub on 30 former grazing parcels in the Santa Barbara foothills over the past 12 years. At each site, ecologists estimated the percent cover of native plants and recorded how many years had passed since restoration began. The data are in a data frame called `sites`.

Variable	Description
`native_cover`	Native plant cover (%)
`years`	Years since restoration began (0 to 12)

Equation (given)

$$\text{native_cover} = 12.4 + 4.8 \cdot \text{years} + \varepsilon, \qquad \varepsilon \sim \text{Normal}(0,\ 35)$$

Visual



Computational

```
Call:
lm(formula = native_cover ~ years, data = sites)

Residuals:
    Min      1Q  Median      3Q     Max
-12.684  -3.935   1.251   4.372   8.158

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  11.4045     2.5373   4.495  0.00011 ***
years         4.7494     0.3553  13.368 1.12e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

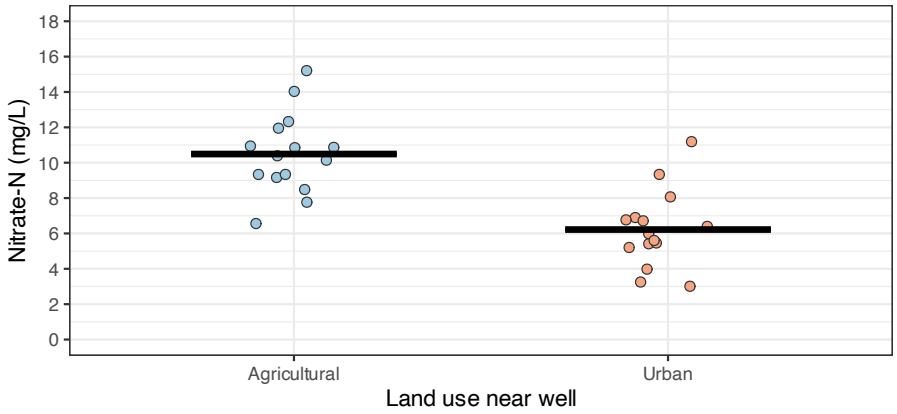
Residual standard error: 6.187 on 28 degrees of freedom
Multiple R-squared:  0.8645,    Adjusted R-squared:  0.8597
F-statistic: 178.7 on 1 and 28 DF,  p-value: 1.12e-13
```

2. Nitrate in domestic wells

Nitrate from fertilizer and septic systems is one of the most common groundwater contaminants in California's agricultural valleys. A water district sampled 30 domestic wells, 15 surrounded mostly by farmland and 15 in urban neighborhoods, and measured nitrate in each. The data are in `wells`. In the figure, black bars mark the group means and the dashed line is the federal drinking-water limit.

Variable	Description
`nitrate`	Nitrate-N concentration (mg/L)
`land_use`	Land use near the well: Agricultural (reference level) or Urban

Visual (given)



Equation

$$\text{nitrate} = 10.49 - 4.27 \cdot \text{Urban} + \varepsilon, \qquad \varepsilon \sim \text{Normal}(0,\ 2.22)$$

Urban = 1 for urban wells and 0 otherwise

Computational

```
Call:
lm(formula = nitrate ~ land_use, data = wells)

Residuals:
    Min      1Q  Median      3Q     Max
-3.9308 -1.1542 -0.1663  0.6456  4.9712

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)    10.4908     0.5730  18.310  < 2e-16 ***
land_useUrban  -4.2710     0.8103  -5.271 1.32e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.219 on 28 degrees of freedom
Multiple R-squared:  0.4981,    Adjusted R-squared:  0.4801
F-statistic: 27.78 on 1 and 28 DF,  p-value: 1.322e-05
```

3. Erosion after a wildfire

After a wildfire, burned hillslopes can lose a lot of soil during the first rainy season. Researchers installed silt fences on 36 hillslope plots, 12 in each burn severity class, and weighed the sediment each fence caught over the winter. The data are in `plots`.

Variable	Description
`erosion`	Sediment caught over the first winter (Mg/ha)
`severity`	Burn severity: Low (reference level), Moderate, or High

Computational (given)

```
Call:
lm(formula = erosion ~ severity, data = plots)

Residuals:
    Min      1Q  Median      3Q     Max
-4.0361 -2.0697 -0.1013  2.0080  7.0912

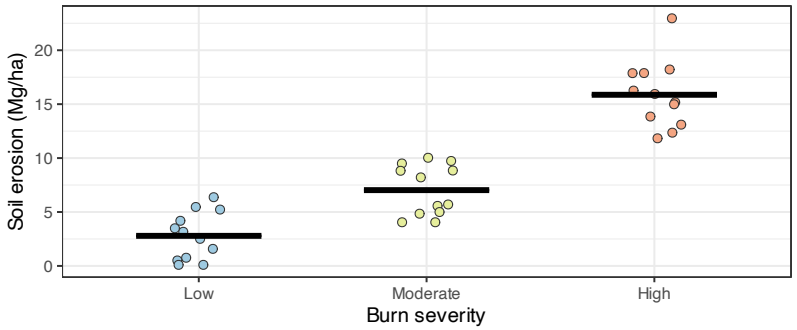
Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)       2.7891     0.7461   3.738 0.000702 ***
severityModerate  4.2401     1.0551   4.019 0.000319 ***
severityHigh     13.0811     1.0551  12.398 5.76e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.584 on 33 degrees of freedom
Multiple R-squared:  0.8291,    Adjusted R-squared:  0.8187
F-statistic: 80.03 on 2 and 33 DF,  p-value: 2.197e-13
```

Equation

$$\text{erosion} = 2.79 + 4.24 \cdot \text{Moderate} + 13.08 \cdot \text{High} + \varepsilon, \qquad \varepsilon \sim \text{Normal}(0,\ 2.58)$$

Visual



4. Do solar trackers pay off?

A community solar program runs 40 identical 10 kW solar arrays. Half are on fixed racks and half are on single-axis trackers that turn to follow the sun. For each array, analysts recorded one day's energy output and that day's solar irradiance. The data are in `arrays`.

Variable	Description
`energy`	Energy produced (kWh/day)
`irradiance`	Solar irradiance (kWh/m^2/day)
`mount`	Fixed (reference level) or Tracker

Computational (given)

```
Call:
lm(formula = energy ~ irradiance + mount, data = arrays)

Residuals:
    Min      1Q  Median      3Q     Max
-6.7249 -1.5375  0.5977  1.9330  3.9247

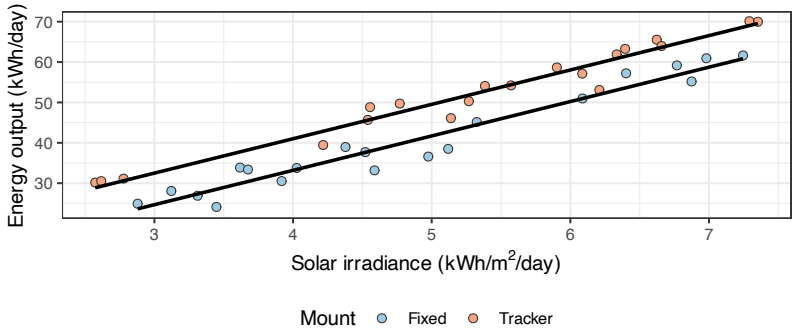
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   -0.8617     1.6202  -0.532    0.598
irradiance     8.5134     0.3087  27.578  < 2e-16 ***
mountTracker   7.8225     0.8727   8.963  8.3e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.725 on 37 degrees of freedom
Multiple R-squared:  0.9623,    Adjusted R-squared:  0.9602
F-statistic: 471.8 on 2 and 37 DF,  p-value: < 2.2e-16
```

Equation

$$\text{energy} = -0.86 + 8.51 \cdot \text{irradiance} + 7.82 \cdot \text{Tracker} + \varepsilon, \qquad \varepsilon \sim \text{Normal}(0,\ 2.72)$$

Visual

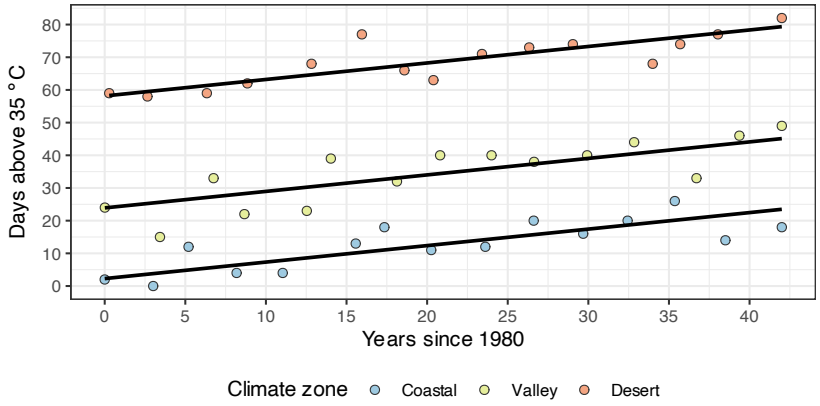


5. Extreme heat days

Extreme heat is becoming more common across California, but how much heat a place already gets depends on where it is. Using weather station records, a climate team counted the number of days each year with a high temperature above 35 °C in three climate zones (15 station-years per zone). The data are in `stations`.

Variable	Description
`heat_days`	Days per year with a high temperature above 35 °C
`years`	Years since 1980
`zone`	Coastal (reference level), Valley, or Desert

Visual (given)



Equation

$$\text{heat_days} = 2.27 + 0.50 \cdot \text{years} + 21.63 \cdot \text{Valley} + 55.88 \cdot \text{Desert} + \varepsilon, \qquad \varepsilon \sim \text{Normal}(0,\ 4.95)$$

Computational

```
Call:
lm(formula = heat_days ~ years + zone, data = stations)

Residuals:
     Min       1Q   Median       3Q      Max
-10.6349  -2.3514   0.0913   3.3714  10.7939

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.2737     1.7400   1.307    0.199
years         0.5048     0.0573   8.811 5.25e-11 ***
zoneValley   21.6307     1.8093  11.955 6.05e-15 ***
zoneDesert   55.8793     1.8092  30.886  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

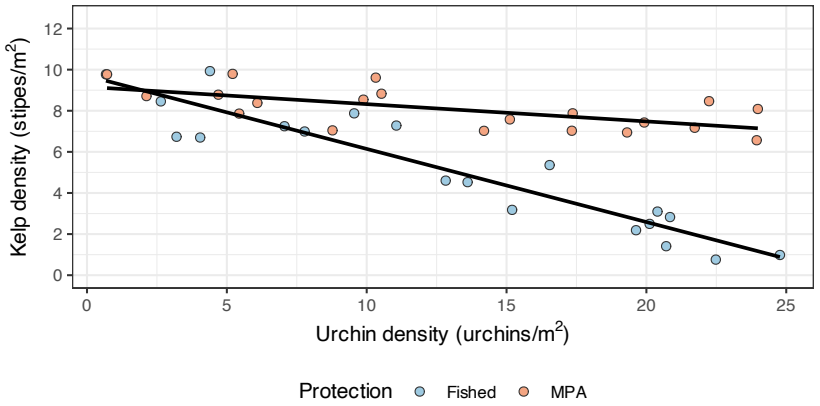
Residual standard error: 4.954 on 41 degrees of freedom
Multiple R-squared:  0.9625,    Adjusted R-squared:  0.9598
F-statistic: 351.2 on 3 and 41 DF,  p-value: < 2.2e-16
```

6. Urchins, kelp, and marine protected areas

Purple sea urchins eat giant kelp, and urchin outbreaks can turn kelp forests into urchin barrens. Inside marine protected areas (MPAs), urchin predators like California sheephead and spiny lobster are more common, which may change how strongly urchins affect kelp. Divers surveyed 40 transects (20 fished, 20 inside MPAs) for urchin and kelp density. The data are in `transects`.

Variable	Description
`kelp`	Kelp density (stipes/m^2)
`urchins`	Purple urchin density (urchins/m^2)
`protection`	Fished (reference level) or MPA

Visual (given)



Equation

$$\text{kelp} = 9.70 - 0.36 \cdot \text{urchins} - 0.54 \cdot \text{MPA} + 0.27 \cdot \text{urchins} \cdot \text{MPA} + \varepsilon, \qquad \varepsilon \sim \text{Normal}(0,\ 0.94)$$

Computational

```
Call:
lm(formula = kelp ~ urchins * protection, data = transects)

Residuals:
     Min       1Q   Median       3Q      Max
-1.82804 -0.58740 -0.05438  0.57571  1.79063

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)          9.70140    0.41933  23.136  < 2e-16 ***
urchins             -0.35575    0.02821 -12.612 8.97e-15 ***
protectionMPA       -0.53827    0.59691  -0.902    0.373
urchins:protectionMPA  0.27170    0.04013   6.770 6.58e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9372 on 36 degrees of freedom
Multiple R-squared:  0.8812,    Adjusted R-squared:  0.8713
F-statistic: 89.04 on 3 and 36 DF,  p-value: < 2.2e-16
```

7. Thinning, rainfall, and tree growth

Forest managers want to know whether thinning crowded conifer stands helps trees grow more in wet years. Across 50 plots in the Sierra Nevada, foresters measured average annual diameter growth, annual precipitation, and stand density. The data are in `trees`.

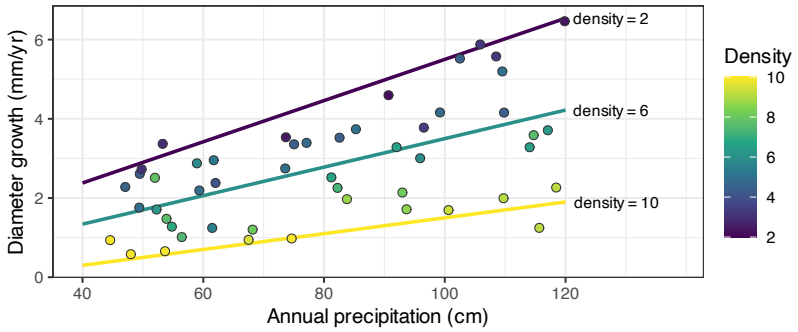
Variable	Description
`growth`	Average annual diameter growth (mm/yr)
`precip`	Annual precipitation (cm, 40 to 120)
`density`	Stand density (hundreds of trees per hectare, 2 to 10)

Equation (given)

$$\text{growth} = 0.5 + 0.06 \cdot \text{precip} - 0.10 \cdot \text{density} - 0.004 \cdot \text{precip} \cdot \text{density} + \varepsilon,$$

$$\varepsilon \sim \text{Normal}(0,\ 0.25)$$

Visual



Computational

```
Call:
lm(formula = growth ~ precip * density, data = trees)

Residuals:
     Min       1Q   Median       3Q      Max
-1.11459 -0.29709 -0.04307  0.26111  1.15325

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)    -0.009053   0.697602  -0.013    0.990
precip          0.068203   0.008218   8.300 1.07e-10 ***
density         0.020232   0.102761   0.197    0.845
precip:density -0.005754   0.001218  -4.725 2.20e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4857 on 46 degrees of freedom
Multiple R-squared:  0.8903,    Adjusted R-squared:  0.8831
F-statistic: 124.4 on 3 and 46 DF,  p-value: < 2.2e-16
```